The University of Nottingham

SCHOOL OF Computer Science

A LEVEL THREE MODULE, SPRING SEMESTER 2018-2019

Computer Vision

Time allowed TWO HOURS

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

Answer THREE Out of FOUR Questions

Each question carries 20 marks

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

ADDITIONAL MATERIAL: **None**

INFORMATION FOR INVIGILATORS: **None**

Marking Scheme

K: knowledge
C: Comprehension
P: problem solving

1. The following questions are pertaining to Features, Object Recognition, and Convolutional Neural Networks.

a. Scale Invariant Feature Transform (SIFT) consists of four main components, i.e., scale-space extrema detection, keypoint localisation, orientation assignment, and keypoint descriptors.

i. Define the terms "scale-space" and "scale-space extrema".

[2 Marks]

Answer: (K)
The original image is convolved with a series of Gaussian functions with increasing standard deviations to obtain scale space images. The higher the standard deviation is, the larger the scale is. [1 mark]

Scale-space extrema are the maximum or minimum values within a neighbourhood of a pixel along the spatial and scale dimensions. [1 mark]

ii. To locate a keypoint, the sum of squared differences (SSD) plays a central role. The formula to calculate SSD of a local image window is given as

$$E(u,v) = \sum_{(x,y)\in W} \big(I(x+u, y+v) - I(x,y)\big)^2$$

where $I(x,y)$ is the image pixel value at coordinates (x,y), and (u,v) the vector representing a small shift of image window W. Clearly, SSD varies with (u,v). Explain conceptually how (u,v) is used, in relation to SSD, to determine a keypoint.

[2 Marks]

Answer: (K,C)
Placing a window centred on a keypoint, it is assumed that moving this window in all directions would cause large SSD. [1 mark]

To find which direction (u,v) would cause the least SSD is the same as finding the eigenvector that corresponds to smallest eigen value of the Hessian matrix of the image window. For a keypoint, this smallest eigen value should be large enough. [1 mark]

iii. What is the purpose of orientation assignment of the local patch centred at a keypoint? Explain, with the example in Figure 1, the possible consequence of omitting the step of orientation assignment.

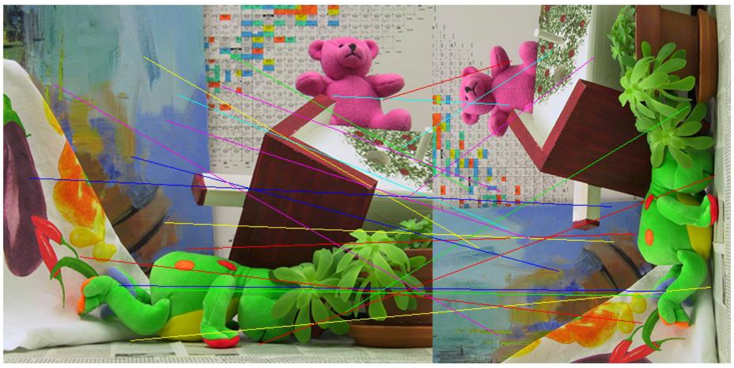

Image 1 Image 2

Figure 1

[2 Marks]

Answer: (C)
The orientation of a keypoint is generally aligned with the x-axis so that the final SIFT descriptor is easily obtained and not affected by the orientation of the keypoint in an image. [1 mark]

In Figure 1, the image 2 is the result of image 1 being rotated 90 degrees and scaled down. If omitting the orientation assignment step, the SIFT descriptor obtained in image 1 will not match with the ones in image 2, as the histogram of the oriented gradient in the descriptor is sensitive to rotation change. [1 mark]

iv. Describe the creation of the keypoint descriptor, including a justification of its main components.

[4 marks]

Answer: (K, C)
Take 16x16 square window around detected feature. Compute edge orientation (angle of the gradient) for each pixel. Throw out weak edges (threshold gradient magnitude). Create histogram of surviving edge orientations. [2 marks]

Histogram is generally translation invariant if the translation is smaller than the size of the image cell for histogram. [1 mark]

Different image cells record the spatial information and therefore concatenating the histograms of these cells would keep the spatial arrangement. [1 mark]

b. Feature representation plays a vital role for object classification, e.g., to classify whether an image contains a cat or not.

(a) (b)

Figure 2

i. Briefly describe the concept of Bag of Features and illustrate the similarity and difference between the bag of features of image (a) and image (b) in Figure 2.

[4 Marks]

Answer: (K, C)
First, take a bunch of images, extract features, and build up a "dictionary" or "visual vocabulary" – a list of common features. [1 mark]
Given a new image, extract features. For each feature, find the closest visual word in the dictionary. Build a histogram to represent the image [1 mark]

Image a and b would both have features that represent the white furry texture but with image a having higher count of such feature and image having lower count. [1 mark]

Image a has features that would differentiate cat from non-cat, e.g., eyes. [1 mark]

ii. Image features can also be captured by a Convolutional Neural Network (CNN) that is trained to classify the images. Explain how features can be learned by a CNN.

[4 marks]

Answer: (K, C)
CNN would build an end-to-end framework to learn image features as well as classifying them into different categories. [1 mark]

In the early (lower) convolutional layers, common features are extracted. [1 mark]

In the later (higher) convolutional layers, label specific features will be formed. [1 mark]

In the full connected layers, a classifier is learned. [1 mark]

iii. Assume that you have 100 samples of cat images and 100 samples of non-cat images. You can either use Bag of Features of these samples to train a linear Support Vector Machine (SVM), or use original images to train a CNN, to classify the images in Figure 2. Which method would you prefer to choose? Justify your answer.

[2 Marks]

Answer: (C,P)
Since the number of training samples are very small [1 mark], if no other information is available, Bag of features + SVM would be preferred. [1 mark]

Alternative answer may be acceptable if one discusses that pretrained CNN may be fine-tuned for the small sample size problem here.

2. The following questions are pertaining to Stereo Vision, Motion, and Optical Flow.

a. Disparity Gradient (DG) constraint has been exploited in several Stereo Vision algorithms to find the correspondence between pixels in left and right images.

i. Explain the terms "Disparity Gradient" and "Disparity Gradient constraint".

[3 marks]

Answer: (K)
Disparity gradient is defined as the ratio of the difference between disparities of two neighbouring pixels over the distance of the two pixels in cyclopean image. [2 marks]

Disparity gradient constraint says that it should be less than a certain limit. According to the results of psychophysical experiments this limit is 1. [1 mark]

ii. Explain how the Disparity Gradient constraint was incorporated in the PMF algorithm.

[3 marks]

Answer: (K, C)
PMF incorporate DG constraint into a neighbourhood support algorithm, i.e., neighbouring matches exchange support if they satisfy a DG constraint. [1 mark]

Build data structures representing possible matches; include the two primitives plus a matching strength. [1 mark]

Support(M,Ni) = Strength(Ni)/Separation(M,Ni), the probability of matches satisfying DG by accident increases with their separation. [1 mark]

iii. In Zhang and Shan's progressive scheme, the Disparity Gradient constraint is translated to the disparity smoothness within a neighbourhood of pixels. Describe conceptually how pixels with known disparities within the neighbourhood of a pixel 'A' are used to determine the disparity uncertainty of the pixel 'A'.

[4 marks]

Answer: (K, C)
Nearby points should have similar disparities, if there is already a match near 'A' we expect 'A' to have the same disparity, plus some additive Gaussian variation (like image noise). [2 marks]

Each nearby match generates a separate prediction of d for 'A'. The spread of these predictions across the image gives a measure of uncertainty. [2 marks]

b. Both Horn-Schunck and Lucas Kanade algorithms are designed for optical flow problems.

i. Explain why Horn-Schunck is considered a global algorithm whilst Lucas Kanade a local algorithm.

[2 marks]

Answer: (K, C)
The objective function of Horn-Schunck algorithm is formed by summing over all image pixels of their brightness constancy and smoothness constraint. [1 mark]

For Lucas Kanade, each local image window has its own objective function which is simply a least square between predicated image by applying the same optical flow vector for all pixels within the window in the current time t, and the actual image in the next time t+1. [1 mark]

ii. Describe what an image pyramid is. Using the Horn-Schunck algorithm as an example, explain how image pyramid can help speed up the optical flow algorithm.

[4 marks]

Answer: (K, C)
A series of images. Each is smaller than the last (usually half the spatial resolution along each axis)
Each pixel in a smaller image corresponds to a number of pixels (usually 4) in the larger one. Images are often blurred before being reduced [2 marks]

Horn-Schunck is an iterative algorithm. Using image pyramid, optical flow can be obtained at the lower-resolution layers of image pyramid and then passed to lower layers as an initialisation for fine tuning. [2 marks]

c. Notice that both stereo vision and motion analysis involve directly and/or indirectly finding correspondences between two images, to obtain disparity for stereo vision problem and optical flow for motion analysis, respectively. Discuss whether it is possible to apply optical flow algorithms (e.g., Horn-Schunck or Lucas Kanade) to find disparity map for a pair of stereo images.

[4 marks]

Answer: (C, P)
This is more of an open question. Students are expected to analyse the constraints or assumptions that are made in typical optical flow algorithms, including brightness constancy and smoothness/small motion. [2 marks on such analysis]

Analysis on whether images in stereo vision problems can generally meet these assumptions. If not, what issues may arise if applying optical flow algorithms. [2 marks on this part of the analysis]

3. The following questions are pertaining to Tracking and Segmentation.

a. CONDENSATION is an effective algorithm for real tracking. It implements particle filters that are designed for multimodal tracking.

i. Explain what multimodal tracking means.

[2 marks]

Answer: (K)
In real tracking, there are often several competing hypotheses. [1 mark]

We can't always tell which hypothesis is correct right away and need to maintain multiple hypotheses. [1 mark]

ii. Describe what each particle represents in CONDENSATION algorithm.

[2 marks]

Answer: (K)
Each particle is a pair (x_t,∏_t).
X_t is a hypothesised state value (containing x, y, u, v etc) [1 mark]
∏_t = P(z_t | x_t) is a measure of its fit to the image data, where z_t is the image data at time t. [1 mark]

iii. In CONDENSATION a particle is filtered through the process of tracking. In Figure 3 below, two particles are placed in frame (a), marked by 'A' and 'B', for tracking the person throwing discus. Assuming that at this point the tracker has been running for a while, particle 'A' is more likely to be filtered out than particle 'B' in (b) and (c). Explain why.

[6 marks]

(a) (b) (c)

Figure 3

Answer: (C, P)
This is a somewhat open question. A model solution is given below. However, any similar discussions will be awarded marks accordingly.

Recognise that the 'A' at the upper left corner locates on a person who's walking to the right (suggested by b and c).[0.5 mark]
The 'B' marked on the athlete is the particle that is supposed to be maintained. [0.5 mark]

CONDENSATION projects both particles along a motion that is suitable to depict the motion of discus throw plus some uncertainties. [1 mark]
According to the sequence, the motion of throwing discus is likely to be rotating around y-axis counter clockwise and translating slightly towards lower part of the image along y-axis. [2 marks]
Thus the upper left 'x' will likely be projected to the location that is not coinciding with the person who's walking towards right (i.e., along x-axis) in image (b) and (c). So in (b) and (c) it will likely be filtered out, i.e., disappear. [1 mark]
The particle on the athlete is likely to be maintained and possibly duplicated in (b) and (c). [1 mark]

b. Active Contours are deformable models that can integrate segmentation and tracking for accurate object shape tracking. The modelling of Active Contour consists of internal energy and external energy.

i. Describe the key components of the internal energy of an Active Contour. You do not have to use equations and can illustrate your description with sketches.

[2 marks]

Answer: (K)
Typically two components: continuity and curvature.
Continuity tries to preserve the spacing between points on the contour. [1 mark]
Curvature attempts maintain the smoothness and flatness of the contour. [1 mark]

ii. The external energy usually attracts the contour towards image features such as higher image gradients. Explain why such external energy is important.

[2 marks]

Answer: (K)
Snake is often to capture the edge of an object or a subject that may deform over time. [1 mark]
Higher image gradients correspond to edges. [1 mark]

iii. An Active Contour is initialised on the image in Figure 4. An 8-neighbourhood search window is placed for each of the six key landmarks. Propose a possible set of positions of these six key landmarks and evaluate whether the set of positions would have higher or lower internal and external energy compared to the initial contour. You can illustrate your answer with sketches.

[6 marks]

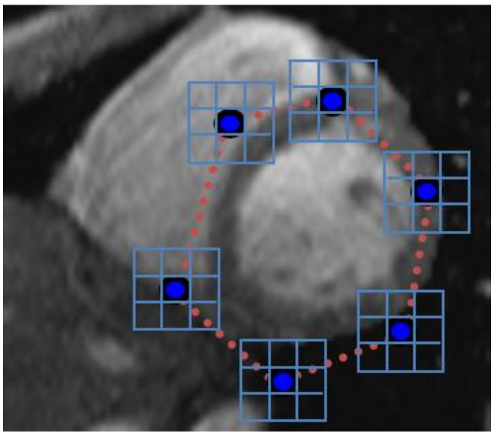

Figure 4

Answer: (C,P)
This is a somewhat open question. Students are expected to provide possible next location of each of the six key points in the image in Figure 4. [2 marks would be given for locating next position in a feasible space]

There should then be discussions and evaluations as to how the external energy look like at those locations [2 marks] and internal energy at those locations [2 marks].

Marks would only be awarded in full if the discussion are consistent with the improvised next locations of key points.

4. The following questions are pertaining to Object Recognition and Evaluation Methods.

a. Viola-Jones object detection framework provides competitive object detection in real time. It was originally designed for face detection. It involves mainly three contributions including integral image calculation for fast feature extraction, boosting for feature selection, and attentional cascade for fast rejection of non-face windows.

i. Explain what an integral image is and how it can be computed efficiently in one pass through the image. You can sketch to help your explanation.

[3 marks]

Answer: (K)
The integral image computes a value at each pixel (x,y) that is the sum of the pixel values above and to the left of (x,y), inclusive. [1 mark]

This can quickly be computed in one pass through the image
Cumulative row sum: $s(x, y) = s(x-1, y) + i(x, y)$ [1 mark]
Integral image: $ii(x, y) = ii(x, y-1) + s(x, y)$ [1 mark]

ii. Boosting is a type of Machine Learning algorithms that combines weak classifiers to form a strong classifier. Explain how boosting is used to select relevant features in Viola-Jones object detection.

[6 marks]

Answer: (K,C)
At each stage of boosting, given reweighted data from previous stage [1 mark]
Train all K single-feature perceptrons [1 mark]
Select the single best classifier at this stage [1 mark]
Combine it with the other previously selected classifiers [1 mark]
Reweight the data [1 mark]
Learn all K classifiers again, select the best, combine, reweight
Repeat until you have T classifiers selected [1 mark]

iii. To evaluate the performance of the Viola-Jones method for face detection, one can make use of precision and recall. Explain what precision and recall means and how they can be used to build Receiver Operating Characteristics (ROC) curves.

[5 marks]

Answer: (K,C)
Precision = TP/(TP+FP) // fraction of responses that were correct [1 mark]
Recall = TP/(TP+FN) // fraction of correct classifications that were identified [1 mark]

Plot of precision against recall as some parameter is varied. [1 mark]
Increasing threshold of the parameter imposes a tighter requirement on matching, which reduces FP, but increases FN. So Precision goes up, but Recall goes down. [2 marks]

b. Ground truth is often a popular reference for the evaluation of stereo vision methods. Explain what ground truth means and give one example of the ground truth for stereo vision problem.

[2 marks]

Answer: (K)

True data as to what is to be measured or predicted. [1 mark]
Ground truth disparity maps produced by calibrating cameras and laser scanner. Laser-measured depth converted to disparity. [1 mark]

c. To evaluate how well an optical flow algorithm works, one can apply the optical flow obtained to the image at time t to predict the image at time $t+1$, which can then be compared with the real image at time $t+1$. Given a 4-by-4 image patch with intensity values at time t in Figure 5(a) and their corresponding optical flow vectors in Figure 5(b), sketch the prediction of the 4-by-4 image patch at time $t+1$. Explain your sketch.

[4 marks]

50	60	65	50
52	80	81	80
82	81	81	81
81	80	80	50

(a)

(0,0)	(0,0)	(0,0)	(0,0)
(0,0)	(1,1)	(1,1)	(1,1)
(1,0)	(1,1)	(1,1)	(1,0)
(1,1)	(1,2)	(1,2)	(0,0)

(b)

Figure 5

Answer: (K,P)

50	60	65	50
52			
	82	80	81
		81	81

[2 marks for the correct numbers at correct locations]

The first two rows are due to no-motion (for 50,60,65,50,52) and blank pixels can either be left blank or filled in with some interpolated value according to smoothness criteria. [1 mark]

The last two rows are either due to the motion (i.e., optical flow) or blank pixels to be filled in with some interpolated value (but can be left blank here). [1 mark]